

papers (Dhuliawala et al., 2023, Madaan et al., 2023, Weller et al., 2023, Weston and Sukhbaatar, 2023, Yasunaga et al., 2024, Zheng et al., 2024) into our desired idea format. For retrieval augmentation, we randomly select $k = 10$ papers from the top-ranked retrieved papers and concatenate their titles and abstracts to prepend to the idea generation prompt. We also append the titles of all previously generated ideas to the prompt to explicitly ask the LLM to avoid repetitions.

To remove duplicated ideas from this large pool of candidate ideas, we first perform a round of deduplication by encoding all seed ideas with `all-MiniLM-L6-v2` from Sentence-Transformers (Reimers and Gurevych, 2020) and then computing pairwise cosine similarities. We set a similarity threshold of 0.8 for the idea deduplication based on manual inspection.⁴ This leaves about 5% non-duplicated ideas out of all the generated seed ideas. We expand more on this duplication issue later in Section 7.1.

3.3 Idea Ranking

The next step is for our ideation agent to rank all the remaining ideas so that we can find the best ones among them. To build such an automatic idea ranker, we use public review data as a proxy. Specifically, we scraped 1200 ICLR 2024 submissions related to LLMs (with keyword filtering) along with their review scores and acceptance decisions. We explored multiple ways of predicting the scores and decisions of these submissions and found that LLMs are poorly calibrated when asked directly to predict the final scores or decisions, but can achieve non-trivial accuracy when asked to judge which paper is better in pairwise comparisons.

We converted the ICLR submissions into our standard project proposal format and randomly paired up accepted and rejected papers and asked LLMs to predict which one is accepted. On this task, Claude-3.5-Sonnet achieves an accuracy of 71.4% with zero-shot prompting. For comparison, GPT-4o achieves 61.1% and Claude-3-Opus achieves 63.5%, and we do not observe significant gains from additional prompting techniques like few-shot or chain-of-thought prompting. We therefore choose the Claude-3.5-Sonnet zero-shot ranker.

In order to obtain reliable scores for all project proposals based on pairwise comparisons, we adopt a Swiss system tournament where all project proposals are paired with those whose accumulated scores are similar, and if the proposals are judged to be better, they gain an additional point. We repeat this for N rounds so the total score of each project proposal will be within the $[0, N]$ range. As a sanity check, we use the Claude-3.5-Sonnet ranker to rank the 1.2K ICLR LLM-related submissions and compare the average review scores of the top 10 ranked papers and the bottom 10 ranked papers in Table 1. We see a clear separation between the top and bottom ranked papers, indicating the effectiveness of the LLM ranker. We choose $N = 5$ for all our experiments since it gives the best ranking result on this validation set. The top-ranked project proposals from the agent will be directly used for the `AI Ideas` condition of the human study.

Since our AI ranker is still far from perfect, we also introduce another experiment condition where the first author of this paper manually reranked the generated project proposals instead of relying on the LLM ranker, and we call this the `AI Ideas + Human Rerank` condition. As we show in

N	Top-10	Bottom-10	Gap
1	6.28	5.72	0.56
2	6.14	5.24	0.90
3	5.83	4.86	0.97
4	5.94	4.99	0.95
5	6.42	4.69	1.73
6	6.11	4.81	1.30

Table 1: Average ICLR review scores of top- and bottom-10 papers ranked by our LLM ranker, with different rounds (N) of pairwise comparisons.

⁴We provide randomly sampled idea pairs and their similarities in Appendix H. We also provide additional implementation details about the ideation agent in Appendix F.

Table 12, 17 out of the 49 ideas in the AI Ideas + Human Rerank condition overlap with the AI Ideas condition, while the other 32 are different, indicating the discrepancy between the LLM ranker and the human expert reranking.

4 Expert Idea Writing and Reviewing

In this section, we shift focus to the human branch of idea generation comparison. We present the details of our human study, including information about the recruited experts, the human idea generation task, and the subsequent review process.

4.1 Expert Recruitment

We recruit our expert participants (including for idea writing and reviewing) by sending sign-up forms to several channels, including: 1) the OpenNLP Slack channel with 1426 NLP researchers from 71 institutions (with consent from the channel manager); 2) Twitter (X); 3) Slack channels of various NLP groups by direct communication with the group members; and 4) official chat app of the NAACL 2024 conference. We also conducted in-person recruitment by giving out name cards and wearing T-shirts⁵ with sign-up links at the NAACL 2024 conference as well as various other local NLP social events. Our study has been approved by the Stanford IRB (ID 74246).

We performed screening on all the US participants⁶ based on their provided Google Scholar profiles. We set a minimum requirement of having published at least one paper at a major AI venue.⁷ We reached out to all participants who satisfied this requirement with the consent form and followed up with the annotation documents for those who consented to participate.

In the end, we recruited $N = 49$ experts for writing ideas, and $N = 79$ experts for reviewing ideas. Note that 24 out of the 79 reviewers also participated in the idea writing, and we made sure no reviewer would review their own idea. This results in $N = 104$ total participants across the two tasks. Each idea writer is asked to write one idea within 10 days and we compensate \$300 for each, with a \$1000 bonus for the top 5 ideas as scored by the expert reviewers. Each idea reviewer is assigned 2 to 7 ideas to review and we collected $N = 298$ unique reviews in total. They are given one week to finish the reviews and we compensated \$25 for each review written by the idea reviewers.

4.2 Expert Qualifications

Our pool of participants is highly qualified and diverse. The 49 idea writers come from 26 different institutions (Table 15) and the majority of them are current PhD students (Figure 3 left). The 79 reviewers come from 32 institutions (Table 16) and are mostly PhD students and Postdocs (Figure 3 right). We use their Google Scholar profiles to extract several proxy metrics, including the number of papers, citations, h-index, and i10-index at the time of their submission. Table 2 shows that our idea writers have an average of 12 papers and 477 citations, while every reviewer has published at least two papers and has an average citation of 635 and h-index of 7. Moreover, based on their survey

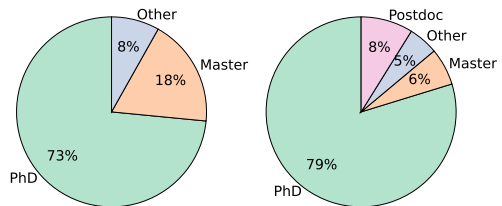


Figure 3: Positions of our idea writer (left) and reviewer (right) participants.

⁵<https://x.com/ChengleiSi/status/1804273510656749649>

⁶We have to recruit participants located in the US due to logistical reasons.

⁷E.g., *ACL, NeurIPS, ICLR, ICML, AAAI.

Idea Writing Participants (N=49)						Idea Reviewing Participants (N=79)				
Metric	Mean	Median	Min	Max	SD	Mean	Median	Min	Max	SD
papers	12	10	2	52	9	15	13	2	52	10
citations	477	125	2	4553	861	635	327	0	7276	989
h-index	5	4	1	21	4	7	7	0	21	4
i10-index	5	4	0	32	6	7	5	0	32	6

Table 2: Research profile metrics of the idea writing and reviewing participants. Data are extracted from Google Scholar at the time of idea or review submission.

Metric	Mean	Median	Min	Max	SD
Human Ideas					
Familiarity (1-5)	3.7	4.0	1.0	5.0	1.0
Difficulty (1-5)	3.0	3.0	1.0	5.0	0.7
Time (Hours)	5.5	5.0	2.0	15.0	2.7
Length (Words)	901.7	876.0	444.0	1704.0	253.5
AI Ideas					
Length (Words)	1186.3	1158.0	706.0	1745.0	233.7
AI + Human Rerank Ideas					
Length (Words)	1174.0	1166.0	706.0	1708.0	211.0

Table 3: Statistics of the 49 ideas from each condition.

responses, 72 out of the 79 reviewers have previously reviewed for major AI conferences or journals. These statistics indicate that our participants are highly qualified and have substantial research experience.⁸

4.3 Idea Writing

We report statistics of our idea writers’ ideas to measure their quality. As shown in Table 3, idea writers indicate a moderately high familiarity with their selected topic (3.7 on a 1 to 5 scale), and indicate the task as moderately difficult (3 on a 1 to 5 scale). They spent an average of 5.5 hours on the task and their ideas are 902 words long on average. These indicate that participants are putting substantial effort into this task.⁹ We also show the distribution of their selected topics in Table 4.

Topic	Count
Bias	4
Coding	9
Safety	5
Multilingual	10
Factuality	11
Math	4
Uncertainty	6
Total	49

4.4 Idea Reviewing

Review Assignment We let all reviewer participants select their top two preferred topics as well as their preferred reviewing load (from 2 to 7). We then randomly assign them to ideas within their selected topics and all ideas are anonymized. In the assignment, we balance the number of ideas from each condition for each reviewer and ensure that each reviewer gets at least one human idea and one AI idea. Every idea is reviewed by 2 to 4 different reviewers. We also avoid assigning ideas written by authors from the same institution to avoid any potential contamination. Table 5 shows that each reviewer wrote an average of 3.8 reviews from 2 or 3 conditions, across 1 to 3 topics.

Table 4: Idea topic distribution.

⁸Detailed breakdown of participant positions is in Appendix K.

⁹See Appendix J for more details on the quality control of human ideas.

Metric	Mean	Median	Min	Max	SD
Ours					
Familiarity (1-5)	3.7	3.0	1.0	5.0	0.9
Confidence (1-5)	3.7	4.0	1.0	5.0	0.7
Time (Minutes)	31.7	30.0	5.0	120.0	16.8
Length (Word)	231.9	208.0	41.0	771.0	112.1
ICLR 2024					
Confidence (1-5)	3.7	4.0	1.0	5.0	0.8
Length (Word)	421.5	360.0	14.0	2426.0	236.4
Length (Word; Strengths & Weaknesses)	247.4	207.0	2.0	2010.0	176.4

Table 6: Statistics of our collected reviews, with ICLR 2024 reviews as a baseline (for the 1.2K submissions that mentioned the keyword “language models”).

Review Quality Check Apart from ensuring reviewer qualifications, we also compute statistics to measure the quality of the reviews in Table 6. On average, the reviewers indicated a familiarity of 3.7 (out of 5) in their selected topic and a confidence of 3.7 (out of 5) in their reviews. This is comparable with the 1.2K ICLR 2024 submissions related to language models, where the reviewers also have an average confidence of 3.7 out of 5. Moreover, reviewers spent an average of 32 minutes on each review, with each review being about 232 words long.

Since our review forms are different from the ICLR review forms, we compare them with the ICLR reviews where we remove the summary and question sections and only count the lengths of the strengths and weaknesses sections. This way, the ICLR reviews have an average length of 247, similar to our collected reviews. As an additional measure of review quality, out of the 298 unique reviews that we have collected, 80 of them provided links to existing papers in their rationales to justify why the proposed method is not novel. These results further validate the high quality of our review data.

Metric	Mean	Min	Max	SD
# Reviews	3.8	2.0	7.0	1.3
# Conditions	2.5	2.0	3.0	0.5
# Topics	1.5	1.0	3.0	0.6

Table 5: Statistics of the review assignment.

5 Main Result: AI Ideas Are Rated More Novel Than Expert Ideas

In this section, we present our main finding on whether LLMs can generate better research ideas than experts. Consistently across three different statistical tests accounting for the possible confounders, we find that AI ideas have higher novelty scores than human ideas while being comparable on all other metrics.

5.1 Test 1: Treating Each Review as an Independent Datapoint

In Test 1, we treat each review as an independent datapoint and aggregate all reviews from the same condition. We treat the `Human Ideas` as the baseline condition and compare it with `AI Ideas` and `AI Ideas + Human Rerank` using two-tailed Welch’s t-tests with Bonferroni correction. We show the barplot in Figure 2 and the detailed numerical results in Table 7. Both `AI Ideas` ($\mu = 5.64 \pm \sigma = 1.76$) and `AI Ideas + Human Rerank` ($\mu = 5.81 \pm \sigma = 1.66$) are significantly better than `Human Ideas` ($\mu = 4.84 \pm \sigma = 1.79$) on the novelty score ($p < 0.01$). In this particular test, the AI ideas in both conditions are also significantly better than human ideas on the excitement score ($p < 0.05$), and the `AI Ideas + Human Rerank` condition is also significantly better than `Human Ideas` in terms of

Condition	Size	Mean	Median	SD	SE	Min	Max	p-value
Novelty Score								
Human Ideas	119	4.84	5	1.79	0.16	1	8	–
AI Ideas	109	5.64	6	1.76	0.17	1	10	0.00**
AI Ideas + Human Rerank	109	5.81	6	1.66	0.16	2	10	0.00***
Excitement Score								
Human Ideas	119	4.55	5	1.89	0.17	1	8	–
AI Ideas	109	5.19	6	1.73	0.17	1	9	0.04*
AI Ideas + Human Rerank	109	5.46	6	1.82	0.17	1	9	0.00**
Feasibility Score								
Human Ideas	119	6.61	7	1.99	0.18	1	10	–
AI Ideas	109	6.34	6	1.88	0.18	2	10	1.00
AI Ideas + Human Rerank	109	6.44	6	1.63	0.16	1	10	1.00
Expected Effectiveness Score								
Human Ideas	119	5.13	5	1.76	0.16	1	8	–
AI Ideas	109	5.47	6	1.58	0.15	1	10	0.67
AI Ideas + Human Rerank	109	5.55	6	1.52	0.15	1	9	0.29
Overall Score								
Human Ideas	119	4.68	5	1.90	0.17	1	9	–
AI Ideas	109	4.85	5	1.70	0.16	1	9	1.00
AI Ideas + Human Rerank	109	5.34	6	1.79	0.17	1	9	0.04*

Table 7: Scores across all conditions by treating each review as an independent datapoint (Test 1). Size is the number of reviews for each condition and the p-values are computed with two-tailed Welch’s t-tests with Bonferroni correction. We **bold** results that are statistically significant (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$). AI ideas are judged as significantly better than human ideas in terms of novelty and excitement while being comparable on all other metrics.

the overall score ($p < 0.05$). We do not observe significant differences between AI-generated ideas and human-written ideas on the other metrics.

5.2 Test 2: Treating Each Idea as an Independent Datapoint

Since we collect multiple reviews for each idea, one could argue that we should not treat each review as an independent datapoint. To account for this potential confounder, we perform Test 2 where we average the scores of each idea and treat each idea as one datapoint. This way, the sample size for every condition will be $N = 49$, namely the number of ideas. We treat the Human Ideas as the baseline condition and compare it with AI Ideas and AI Ideas + Human Rerank using two-tailed Welch’s t-tests with Bonferroni correction. As shown in Table 8, we still see significant results ($p < 0.05$) where both AI Ideas ($\mu = 5.62 \pm \sigma = 1.39$) and AI Ideas + Human Rerank ($\mu = 5.78 \pm \sigma = 1.07$) have higher novelty scores than Human Ideas ($\mu = 4.86 \pm \sigma = 1.26$).

5.3 Test 3: Treating Each Reviewer as an Independent Datapoint

Another possible confounder is that different reviewers might have different biases, for example, some reviewers may be more lenient than others. To account for such reviewer biases, we perform Test